ch12 Gravitation

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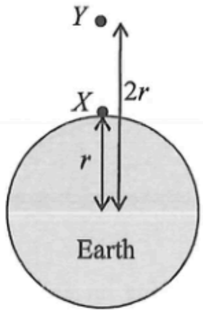
2015 Q11

11. The gravitational force exerted on the Earth by the Sun is F_0. The gravitational force exerted on the Sun by the Earth is

A. equal to F_0 and in the same direction.
B. equal to F_0 and in the opposite direction.
C. much smaller than F_0 and in the same direction.
D. much smaller than F_0 and in the opposite direction.

*12.



Two identical objects located at point X and point Y are at distances r and $2r$ from the Earth's centre. The weight of the object at point X is W. What is the weight of the object at point Y?

A. $0.25W$
B. $0.5W$
C. $0.75W$
D. W

2013 Q15

*15. It is known that the mass of Mars is about $\frac{1}{10}$ of that of the Earth while its radius is about $\frac{1}{2}$ of the Earth's radius. In terms of the gravitational acceleration g on the Earth's surface, the approximate gravitational acceleration on the surface of Mars is

A. $0.2\,g$.
B. $0.4\,g$.
C. $2.5\,g$.
D. $4\,g$.

2018 Q13

*13. A satellite of mass m moves around a planet of mass M in a circular orbit of radius r. What does the angular velocity of the satellite depend on ?

(1) r
(2) m
(3) M

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

2020 Q10

*10. The diameter of Neptune is about 4 times that of the Earth and its mass is about 17 times that of the Earth. Estimate the acceleration due to gravity on Neptune's surface.
Given: acceleration due to gravity on Earth's surface $g = 9.81$ m s^{-2}

A. 2.3 m s^{-2}
B. 9.2 m s^{-2}
C. 10.4 m s^{-2}
D. 41.7 m s^{-2}

2017 Q13

*13. A small object is released from rest at a point very far away from a planet X. The object then starts moving towards X. X does not have an atmosphere. Neglect the effect of other celestial bodies.

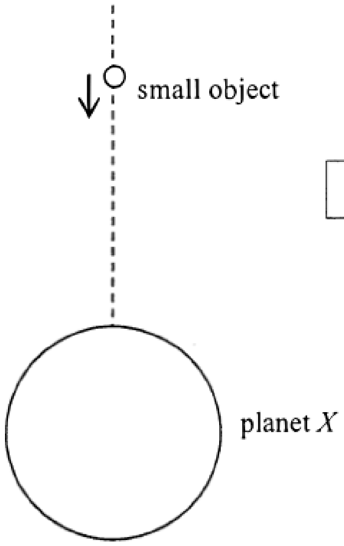
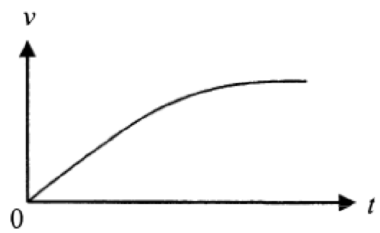


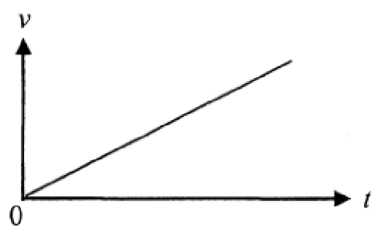
Diagram **NOT** drawn to scale

Which of the following graphs best shows the variation of the velocity v of the object with time t before it hits X?

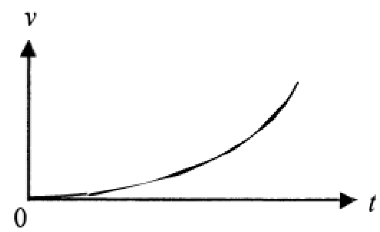
A.



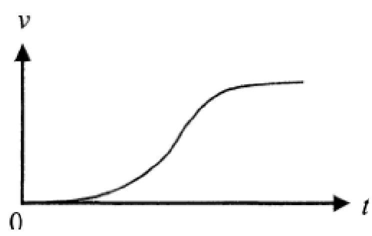
B.



C.

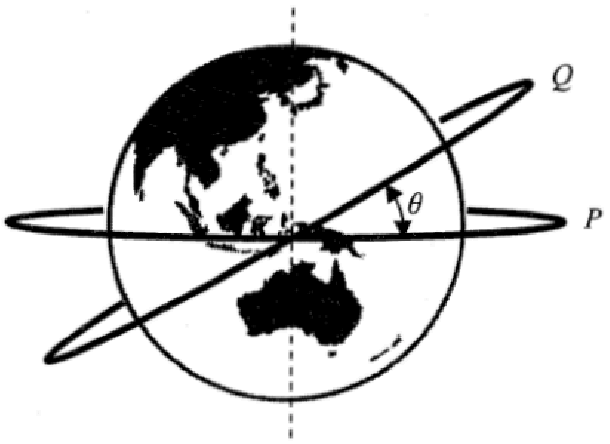


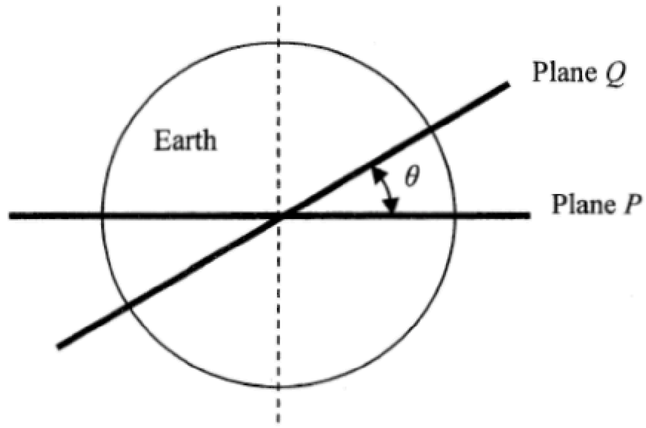
D.



2012 Q14

*14. Two satellites move in circular orbits of the same radius R around the Earth (mass M). The orbits are in two different planes P and Q as shown. Plane P coincides with the Earth's equator while plane Q is inclined to the equator at θ. Which statement is **INCORRECT** ?





A. The speed of satellite P is $\sqrt{\dfrac{GM}{R}}$.

B. The centripetal force acting on satellite Q is pointing along the plane Q.

C. The acceleration of both satellites is the same in magnitude.

D. The period of satellite Q is longer than that of satellite P.

2014 Q12

*12. An artificial satellite revolves in a circular orbit above the Earth's surface at a height equal to the radius of the Earth. Find the acceleration of the satellite in terms of the acceleration due to gravity g on the Earth's surface.

A. $\frac{g}{8}$

B. $\frac{g}{4}$

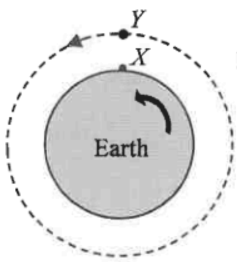
2016 Q14

*14. A satellite orbits the Earth in a circular path of radius 7.2×10^6 m. What is the period of the satellite ?
Given : mass of the Earth = 6.0×10^{24} kg

A. 1.4 hours
B. 1.7 hours
C. 1 day
D. Answer cannot be found as the mass of the satellite is not known.

2022 Q14

*14. In the figure, X is an object resting on the Earth's equator. Y is a geostationary satellite moving in a circular orbit above the equator such that it always appears stationary to an observer on Earth.



NOT drawn to scale

Which descriptions about the motion of X and Y are correct ?

(1) The motion of X and Y takes the same period.
(2) X is moving with a slower speed.
(3) X has a greater acceleration.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2014 Q11

*11. An astronaut inside a spacecraft moving in a circular orbit around the Earth is apparently weightless because

A. the astronaut is too far from the Earth to feel the Earth's gravitational force.
B. the astronaut and the spacecraft are both moving with the same acceleration towards the Earth.
C. the Earth's gravitational force on the astronaut is balanced by the reaction force of the spacecraft's floor.
D. the Earth's gravitational force on the astronaut is balanced by the centripetal force.

*13. A satellite orbits the Earth in a circular orbit of radius r with a period of 24 hours. The period of another satellite which orbits the Earth in a circular orbit of radius $2r$ is

A. less than 24 hours.
B. 24 hours.
C. between 24 hours and 48 hours.
D. more than 48 hours.

2026 Q11

*11. Satellites X and Y revolve around the Earth in circular orbits, in the same plane and in the same direction. The radii of their orbits are r and $4r$ respectively. The figure below shows an instant that X, Y and the Earth are on a straight line.

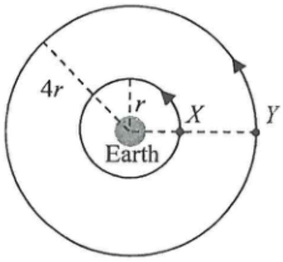
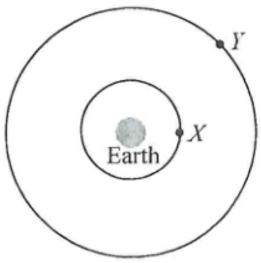


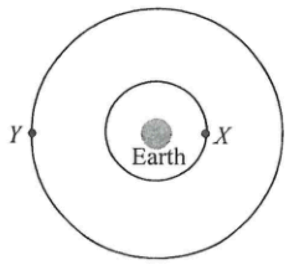
Diagram **NOT** drawn to scale

When X completes one cycle, which of the following diagrams best represents the position of Y ?

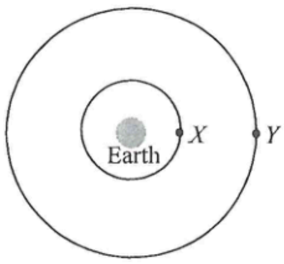
A.



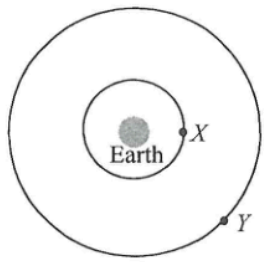
B.



C.



D.



*12. A rocket on the Earth's surface is launched by an upward thrust F, producing an upward acceleration g, where g is the acceleration due to gravity near the Earth's surface, as shown in Figure (a). R is the radius of the Earth.

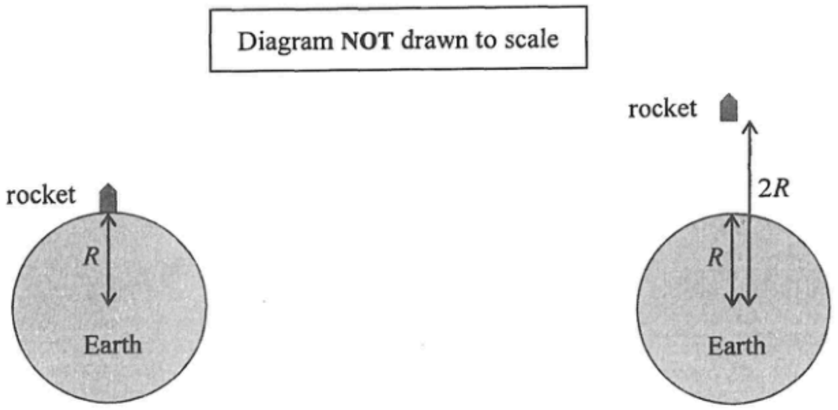


Figure (a) Figure (b)

Figure (b) shows an identical rocket at a distance $2R$ away from the Earth's centre. What is the upward thrust required for producing the same upward acceleration g ? Neglect air resistance.

A. $\frac{F}{4}$

B. $\frac{F}{2}$

C. $\frac{5F}{8}$

D. $\frac{3F}{4}$